

Aiden Slabiak

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EDUCATION

University of Alabama

B.S. Computer Science

December 2026

GPA: 3.7/4.0

COURSEWORK

Relevant Coursework: Operating Systems, Digital Logic, Microcomputers, Algorithms & Data Structures, Software Design & Engineering, Reinforcement Learning

Study Abroad: Tokyo, Vietnam, Taiwan

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript

Systems: Intel VT-x, x86-64, Windows Internals, Multithreading, Winsock/TCP/IP, Linux/Unix

Tools: WinDbg, GDB, Visual Studio, Git, GitHub, Bitbucket, WSL

EXPERIENCE

Software Developer Intern | *Scientific Research Corporation*

May 2026 – Present

- Built a reusable Salesforce Lightning Web Component for CSV ingestion with client-side parsing, header validation, row processing, and error reporting before Apex persistence.
- Developed Apex batch-processing and data-import logic with record-sharing and field-level security enforcement to validate, transform, and insert production data.
- Integrated data-processing components with Salesforce Flow Builder and collaborated through code reviews, testing, Jira, and Bitbucket in an Agile development environment.

Undergraduate Research Assistant | *University of Alabama*

Jan. 2026 – May 2026

- Evaluated AI models through controlled experimentation and performance analysis to characterize system capabilities and limitations.
- Collaborated with faculty and graduate researchers to translate experimental findings into requirements for an AI-assisted educational platform.

PROJECTS

Intel VT-x Hypervisor | *C++, x86-64, Windows Kernel, WinDbg*

Aug. 2026

- Built a Windows x86-64 hypervisor using Intel VT-x, implementing VMXON/VMCS initialization, VM-entry/VM-exit handling, and per-processor virtual CPU state.
- Configured guest/host state and VM-execution controls, including control registers, descriptor tables, CPUID interception, MSR bitmaps, and instruction-pointer advancement.
- Debugged VM-entry failures, invalid architectural state, and repeated VM exits using WinDbg kernel debugging and Intel VMX architectural documentation.

SIMD-Accelerated Matrix Library | *C++, AVX2/AVX-512, Intrinsics*

Mar. 2026

- Implemented SIMD-accelerated matrix operations using AVX2/AVX-512 intrinsics, aligned memory, vectorized computation, and cache-aware loop optimizations.
- Achieved approximately $3.5\times$ speedup over scalar implementations while preserving numerical correctness through automated tests.

6-DOF Flight Dynamics Simulator | *C++, Winsock, React, Three.js*

Jan. 2026

- Developed a multithreaded six-degree-of-freedom flight simulation engine modeling rigid-body motion, vehicle dynamics, aerodynamic forces, and weather.
- Designed a non-blocking Winsock TCP interface to stream simulation data at approximately 60 Hz between the C++ engine and visualization client.
- Built an Electron/React and Three.js visualization interface with real-time telemetry, configurable simulation parameters, and interactive controls.